Coding protocol

Registered Report: Is Algorithm Aversion a Rule or a Weight?

This protocol describes how the interview transcripts are coded. It belongs to the sections "Transcript coding", "Catalogue of strategies and reasons" and "Analysis plan" (Steps 1 and 2) of the Stage 1 manuscript.

1 Attributes and level order

Each choice task shows two automated systems, A and B, described by five attributes with three levels each. The rules below use the level order of this table (0, 1, 2).

Attribute	0	1	2
How quickly do you receive the final decision?	6 weeks	within one week	instantly
What reasons are given?	no reasons given	brief standard reasons	full personal reasons
Applications prioritised during busy periods	families with children	unemployment applications	pension applications
Is there a human review?	no human review	human review of flagged cases	human review for every case
How can you appeal the decision online?	one-click appeal in OmaKela	free-form email appeal	online video call with a caseworker

Faster decisions, more detailed reasons and more human review have higher indices. Priority has no natural order, so each priority level has its own reason. For the appeal channel, personal appeal orders the levels by human contact (video call over email over one-click) and convenient appeal by effort (one-click over email over video call); both orders are assumptions of the design.

2 Catalogue

Every item is rated from 0 to 100. Attribute-linked reasons follow the two-part format of Fulawka, Hertwig and Wulff (2026): what the reason considers and which system it prefers. Each reason has a formal rule that returns +1 (System A), -1 (System B) or 0 (no preference) for any pair of profiles.

ID	Item	Definition	Source
S1	Lexicographic	The respondent describes one characteristic that decided their choices whenever the two systems differed on it; other characteristics mattered only when that one was the same. Prefers the system with the better level of the decisive characteristic; if equal, the next characteristic the respondent names.	Banavar and Bansak (2026)
S2	Compensatory	The respondent describes weighing or counting several characteristics against each other, accepting a worse level on one for a better level on another.	Banavar and Bansak (2026)
R1	Speed	Considers how quickly the final decision arrives. Prefers the faster decision.	Dong et al. (2025)
R2	Explanation	Considers how detailed the reasons given for the decision are. Prefers more detailed reasons.	Mahmud et al. (2022)
R3	Human oversight	Considers how much a human is involved in the decision. Prefers more human involvement.	Mahmud et al. (2022)
R4	Automation preference	Considers how much a human is involved in the decision. Prefers less human involvement.	Logg et al. (2019)
R5	Personal appeal	Considers how the decision can be appealed. Prefers the channel with more human contact.	attribute set
R6	Convenient appeal	Considers how the decision can be appealed. Prefers the channel that takes less effort.	attribute set
R7	Priority for families	Considers which applications are prioritised during busy periods. Prefers priority for families with children.	attribute set
R8	Priority for the unemployed	Same consideration. Prefers priority for unemployment applications.	attribute set
R9	Priority for pensioners	Same consideration. Prefers priority for pension applications.	attribute set
M1	Error intolerance	Mistakes by an automated system are less acceptable than the same mistakes by a person, or a system must not make mistakes.	Dietvorst et al. (2015)
M2	Diminishing sensitivity to error	A system will often be somewhat wrong; near-perfect decisions are what matters, or a person at least sometimes gets it exactly right.	Dietvorst and Bharti (2020)
M3	Desire for control	The respondent wants to be able to intervene, correct, adjust or influence the decision or the process.	Dietvorst et al. (2018)
M4	Subjectivity of the task	Benefit decisions require judgment, empathy or understanding that a system does not have.	Castelo et al. (2019)
M5	Neglect of individual circumstances	A system cannot take the respondent's particular situation, exceptions or special cases into account.	Dietvorst et al. (2015); Mahmud et al. (2022)
M6	Dehumanization	Being decided on by a machine is impersonal, degrading or treats people like numbers.	Dietvorst et al. (2015)
M7	General distrust of automation	The respondent does not trust automated systems, computers or AI in general, without naming a specific feature.	Mahmud et al. (2022)

ID	Item	Definition	Source
M8	Consistency of algorithms	A system treats everyone the same or applies the rules consistently, as a reason for or against it.	Dietvorst et al. (2015)

Mechanisms describe why a respondent might avoid automated decisions in general; they do not distinguish two automated systems and have no formal rule. "Attribute set" marks reasons that follow from the design.

3 Classification

A reason or mechanism counts as present if its rating reaches a threshold selected on the data as in Fulawka, Hertwig and Wulff (2026): for each threshold from 0 to 100 in steps of 10, the reasons at or above it predict each choice by the sign of the sum of their preferences (a random choice if none is identified or the sum is zero), and the highest threshold whose agreement with the actual choices is within one percentage point of the best agreement is used. A respondent states an aversion mechanism if any of M1 to M7 is present; M8 is coded but not counted, because the literature names consistency both as an advantage and as a disadvantage of algorithms.

A respondent is predominantly lexicographic if the S1 rating exceeds the S2 rating, predominantly compensatory in the reverse case, and unclassified if both ratings are below 50 or equal. Lexicographic respondents for whom the coder returns no decisive attribute are also unclassified. As a robustness check, the tests are repeated with a strict rule (one strategy item at or above the threshold, the other below it).

4 Step 1: choice of the coding model

The candidates are the four open-weight models listed in Supplement S2. Before any transcript is coded, each candidate receives the definition of each attribute-linked reason together with randomly drawn pairs of systems and states which system the reason prefers (A, B or neither). The answers are scored against the formal rules. The most accurate model is used; a tie goes to the smaller model.

5 Step 2: transcript coding

The prompt adapts the verbal-report prompt of Fulawka, Hertwig and Wulff (2026). The choice problem is replaced by the attribute table, because one transcript covers all six tasks. One call covers one catalogue item and one transcript. Decoding is deterministic (temperature 0). The rating is read from the line `FINAL ASSESSMENT:`; outputs that cannot be parsed are set to 50 and counted. The prompt, the definitions and the attribute table are in English; transcripts are coded in their original language without translation.

5.1 System message (reasons)

```
You are a decision analyst specializing in evaluating decision-making processes. Your
↪  task is to assess whether a specific decision reason is present in an interview
↪  transcript of an individual who made six choices between two hypothetical
↪  automated systems.

Available information. Each choice was between two systems, A and B, described by five
↪  characteristics with the following possible levels:
```

```
- How quickly do you receive the final decision? 6 weeks; within one week; instantly
- What reasons are given? no reasons given; brief standard reasons; full personal
↪  reasons
- Applications prioritised during busy periods: families with children; unemployment
↪  applications; pension applications
- Is there a human review? no human review; human review of flagged cases; human
↪  review for every case
- How can you appeal the decision online? one-click appeal in OmaKela; free-form email
↪  appeal; online video call with a caseworker

A decision reason specifies which information the individual considers and which
↪  system the reason prefers. A transcript is a written conversation, in retrospect,
↪  between an interviewer and the individual about how the individual made the six
↪  choices. The interviewer's turns are marked INTERVIEWER, the individual's turns
↪  RESPONDENT.

Task. Assess, based on the RESPONDENT turns only, whether the individual used the
↪  reason to make their choices. Proceed step by step.

Step 1: State which characteristic and which levels the reason refers to, and
↪  summarize what information about that characteristic the individual mentions
↪  considering.

Step 2: Evaluate and summarize the individual's justification for their choices. Focus
↪  on the described deliberation process. A stated criterion ("I took the faster
↪  one") counts as information about the process. A bare statement of which system
↪  was chosen ("I chose System A") does not, and must be ignored.

Step 3: Compare the information the reason considers with what the individual
↪  considered, and compare the reason's preference with the individual's
↪  justification. Based on both comparisons, assess whether the individual used the
↪  reason. Give your final assessment on a scale from 0 to 100:
- 0: certain that the reason was not used;
- 50: uncertain whether the reason was used;
- 100: certain that the reason was used.

Output structure. Use exactly this template:
STEPS: <your reasoning for each step>
FINAL ASSESSMENT: <integer from 0 to 100>
```

5.2 User message

```
Decision reason: <definition of the item>

Transcript:
<full transcript with INTERVIEWER / RESPONDENT labels>
```

5.3 System message (aversion mechanisms M1 to M8)

Identical to the message for reasons, with Step 1 replaced by:

```
Step 1: Summarize what the individual says about automated decisions in general,
↪  anywhere in the transcript.
```

5.4 System message (strategy items S1 and S2)

Identical to the message for reasons, with "decision reason" replaced by "decision strategy" and Step 1 replaced by:

```
Step 1: Summarize how the individual describes combining the characteristics: whether
↪  one characteristic decided the choice whenever it differed, whether several
↪  characteristics were weighed or counted against each other, or whether the process
↪  is not described.
```

For S1, the output adds:

```
DECISIVE CHARACTERISTIC: <1-5 or NONE>
PREFERRED: <HIGHER | LOWER | one level name for characteristic 3 | NONE>
SECOND CHARACTERISTIC: <1-5 or NONE>
SECOND PREFERRED: <HIGHER | LOWER | level name | NONE>
```

HIGHER means faster, more detailed reasons, more human review or more human contact in the appeal; LOWER means the opposite.

6 Human coding

Two trained coders independently code a random sample of 300 transcripts, 100 per survey language, before the confirmatory analyses. They do not see the respondents' choices or tap logs and code only what the respondent writes; the interviewer's turns serve only to understand what a short answer refers to.

The coders rate every item on the same 0 to 100 scale as the model (0 certain absent, 50 cannot tell, 100 certain present) and give high ratings only where the transcript clearly supports the item. For S1 they also record the decisive characteristic, the preferred direction or level, and a second characteristic if the respondent names one for ties. They follow these rules:

1. A reason is present if the respondent says they considered the characteristic and their justification points in the reason's direction ("I always took the faster one" is high on speed; "speed did not matter to me" is low).
2. A bare statement of which system was chosen is not information about a reason.
3. The strategy items describe the process, not the content. If a respondent describes both strategies, both are rated.
4. Mechanisms are general views about automated decisions and are coded wherever they appear, including in the answer about the respondent's own life.
5. Nothing is inferred from tone or from what a typical respondent might think.
6. A transcript is flagged if the interviewer names a characteristic before the respondent has mentioned it.

The coders code the first 20 transcripts, discuss disagreements on the definitions (not on specific ratings), and then code the remaining 280 without further contact. We report Cohen's kappa between the coders and F1 scores between each coder and the model.